A Human–Computer Interaction Study of User Experience Challenges in Online Banking Portals

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ABSTRACT

The field of Human-Computer Interaction (HCI) is concerned with the design of user interfaces and systems that are not just usable and effective but also provide a high degree of satisfaction to the users of those systems. In the digital age, online banking portals represent one of the most critical and widely used financial systems, accessed by millions of users daily for transactions, account management, and financial services. This paper presents an investigation into the user interfaces and systems of online banking portals and attempts to analyze how well established Human-Computer Interaction principles are applied in real-world banking environments. The study examines common usability and user interface issues associated with online banking portals based on principles developed by renowned HCI experts such as Jakob Nielsen, Don Norman, and Ben Shneiderman. These issues include cognitive overload during authentication, unclear error messages during failed transactions, lack of feedback during processing, and recall-heavy interaction patterns that significantly affect user satisfaction and trust.

1. PROBLEM STATEMENT

Online banking portals are among the most critical digital systems used by the general public. These portals are accessed by users of all ages and technical backgrounds for high-stakes activities such as fund transfers, bill payments, loan applications, and account management. Despite significant advances in web and mobile technologies, many banking portals continue to suffer from serious usability and user experience issues. Common problems include complex multi-step authentication processes imposing high cognitive load, generic and unhelpful error messages during failed transactions, lack of clear system feedback during fund processing, session timeouts without adequate warning, and poor mobile responsiveness. These issues result in user frustration, repeated errors, transaction abandonment, decreased trust in the banking system, and heightened anxiety particularly for elderly or less tech-savvy users. The absence of consistent user-centered design practices in such high-stakes financial systems highlights the urgent need for systematic usability and UX analysis using established Human–Computer Interaction principles.

2. INTRODUCTION

Human–Computer Interaction (HCI) is an interdisciplinary field concerned with the design, evaluation, and use of interactive computing systems in ways that make interaction efficient, effective, and satisfying for users. As digital systems increasingly mediate financial activities, the scope of HCI in the banking domain has expanded beyond purely functional considerations to include usability, cognition, emotional experience, and user trust. This shift reflects the growing recognition that the quality of interaction between humans and banking systems significantly influences system adoption and user confidence.

Traditionally, banking was conducted through physical branch visits, requiring users to interact with bank personnel in person. With the rise of internet banking, financial institutions transitioned to web-based portals to improve accessibility, speed, and efficiency of financial services. While this transition has enhanced convenience, many such systems were developed with a primary focus on functionality and security rather than user experience and interaction quality.

The evolution of HCI emphasizes a move from system-centered design to user-centered design. Modern UI/UX practices highlight principles such as visibility of system status, meaningful feedback after transactions, error prevention and recovery mechanisms, and reduction of cognitive load during authentication. When these principles are not applied in financial contexts, users are more likely to experience confusion, distrust, anxiety, and transaction failures. As a result, online banking portals provide a compelling and highly relevant context for studying usability and user experience challenges.

In the financial domain, web-based banking portals have become the primary interfaces through which critical services such as fund transfers, bill payments, loan management, and investment tracking are accessed. Among these systems, online banking portals are especially significant as they are accessed by a diverse population including students, professionals, and senior citizens, often under time-sensitive and high-stakes conditions. Consequently, the quality of interaction offered by these portals has a substantial impact on user confidence, stress levels, and overall financial inclusion.

Despite their importance, many online banking portals exhibit poor interface usability due to design limitations. Commonly observed issues include overly complex authentication flows, unclear transaction status indicators, session timeouts without warning, poor error messages during failed transfers, and recall-heavy interactions that impose significant cognitive demand on users. These issues often result in

abandoned transactions, security vulnerability from frustrated users writing down credentials, increased call center load, and elevated user anxiety during financial operations.

Jakob Nielsen identifies five key components of usability: learnability, efficiency, memorability, error handling, and user satisfaction. Online banking portals frequently perform poorly across these dimensions. First-time users struggle with complex registration flows; infrequent users face difficulty recalling multi-step login procedures; and users of all experience levels encounter confusing transaction confirmation mechanisms. Inadequate error messages and weak recovery options further diminish overall usability.

Complementing Nielsen's usability framework, HCI principles proposed by Don Norman emphasize visibility of system status, timely feedback, and minimizing the cognitive gap between user intentions and system responses. Similarly, Ben Shneiderman's interaction guidelines stress consistency, error prevention, and recognition-based interaction over recall. Violations of these principles cause users to lose a sense of control, particularly during time-sensitive financial operations such as fund transfers and bill payments.

The present study conducts an analytical examination of existing online banking portals to identify common usability and user experience issues. Rather than proposing the design of a completely new system, the study focuses on understanding how violations of established interaction design principles contribute to increased cognitive load, higher error rates, and elevated user anxiety. By analyzing real-world banking systems through the lens of fundamental HCI principles, the paper highlights how improved adherence to user-centered design practices can reduce user stress and enhance the overall effectiveness of digital banking interfaces.

2.1 OBJECTIVES OF STUDY

- To identify common usability and user experience issues in online banking portals
- To analyze these issues using established Human–Computer Interaction principles
- To understand the impact of poor UX on user anxiety, transaction errors, and satisfaction
- To examine real-world banking portals through heuristic evaluation, user surveys, and task walkthroughs

3. LITERATURE REVIEW

Online banking and digital financial interfaces have been the focus of numerous research studies recognizing their increasing importance in contemporary financial inclusion and digital transformation. Several researchers have explored these systems, specifically focusing on security, usability, and user experience dimensions of internet banking.

Research published in the International Journal of Human-Computer Studies examined the usability of mobile banking applications and found that authentication complexity was the leading cause of user abandonment. The study highlighted that multi-factor authentication, while improving security, significantly increases cognitive load when poorly designed, particularly affecting older users and those with limited digital literacy.

A study titled 'User Experience in Internet Banking: A Systematic Review' published in the Journal of Internet Banking and Commerce analyzed UX challenges across major banking portals. The research identified that session timeout management, unclear error messaging, and poor transaction feedback were the most frequent usability violations, directly correlating with increased user anxiety and decreased trust in digital banking systems.

Research published by IEEE (2022) on 'Cognitive Load in Digital Financial Services' explored how the mental effort required during online banking tasks affects user performance and error rates. The study found that banking portals requiring users to remember multiple credentials — user IDs, passwords, PINs, and OTPs — violated Nielsen's heuristic of recognition over recall, leading to significantly higher error rates compared to portals with simplified authentication.

A study by the ACM on 'Trust and Usability in Online Banking Interfaces' (2023) examined the relationship between interface usability and user trust. The findings demonstrated that users who experienced poor feedback during transactions, such as no loading indicators or ambiguous confirmation messages, reported significantly lower trust in the banking system, even when transactions were ultimately successful.

More recent works have equally addressed mobile banking accessibility. Studies such as 'HCI Evaluation of Mobile Banking Apps for Senior Users' (2024) focus on how poor font sizing, low contrast, and recall-heavy interactions disproportionately affect elderly users, who represent a growing segment of digital banking users. Another study titled 'Stress by Design: Financial Portal UX and User Anxiety' (2025)

demonstrated that interface failures during financial transactions produced measurable physiological stress responses in users.

Despite these contributions, the current literature largely targets either specific security aspects of banking systems or general usability evaluations without applying comprehensive HCI frameworks. Little research has focused on the systematic heuristic evaluation of online banking portals using combined frameworks from Nielsen, Norman, and Shneiderman, especially in the Indian banking context where digital financial inclusion is rapidly expanding. This gap motivates the current research.

4. METHODOLOGY

The current study employs a mixed-methods approach combining heuristic evaluation and user-based usability assessment. The research method is grounded in the analysis of user experience problems encountered while using online banking portals, conducted without the aim of proposing a complete interface redesign.

4.1 STUDY CONTEXT AND SYSTEM SELECTION

The research focuses on existing online banking portals that are publicly accessible and widely used by individuals for financial transactions. Most of these portals require multi-step authentication including User ID, Password, OTP, and sometimes CAPTCHA to access banking services. Such portals were selected because of the high stakes involved, the diverse user base spanning multiple age groups and technical literacy levels, and the critical nature of the tasks performed including fund transfers, bill payments, and account management.

4.2 HEURISTIC EVALUATION

A heuristic evaluation was conducted to identify usability issues based on established Human-Computer Interaction principles.

4.2.1 EXPERT SELECTION

Three evaluators with prior knowledge of Human-Computer Interaction and usability principles participated in the evaluation. Each expert independently reviewed the selected online banking portals to minimize bias and ensure comprehensive identification of usability issues.

4.2.2 EVALUATION CRITERIA

The evaluation was conducted using the following established frameworks:

- Jakob Nielsen's usability heuristics
- Interaction principles proposed by Don Norman
- Interaction guidelines defined by Ben Shneiderman

The experts assessed the portals with respect to visibility of system status, feedback during transactions, error prevention and recovery, consistency, cognitive load, and recognition-based versus recall-based interaction.

4.2.3 OUTPUT OF HEURISTIC EVALUATION

Identified usability problems were documented and categorized according to the violated principles. The potential impact of each issue on user errors, confusion, and stress levels was noted. Usability issues reported by multiple evaluators were considered significant and prioritized for analysis.

4.3 TASK WALKTHROUGH ANALYSIS

To understand user interaction flow in banking portals, a task walkthrough method was employed.

4.3.1 DEFINED TASKS

The following core tasks performed by banking portal users were analyzed:

- Logging into the banking portal using multi-step authentication
- Checking account balance and recent transactions
- Performing a fund transfer to a new beneficiary
- Paying utility bills through the portal
- Downloading account statements

4.3.2 WALKTHROUGH PROCESS

Each task was broken down into sequential steps. At every step, the evaluators examined availability of system feedback, clarity of instructions, presence and usefulness of error messages, error recovery options, and cognitive effort required to proceed. Points where users were likely to experience confusion, delays, or errors were identified and documented.

4.4 USER BASED USABILITY SURVEY

A user survey was conducted to capture real user experiences and perceptions of online banking portals.

4.4.1 SURVEY DESIGN

The survey was designed using Google Forms and included both closed-ended and open-ended questions. The questionnaire focused on ease of login and authentication, clarity of transaction steps, error handling and feedback, cognitive load during operations, stress experienced during banking activities, and overall user satisfaction.

4.4.2 PARTICIPANTS

The study involved two groups of participants:

Usability Experts:

Three evaluators with prior knowledge of Human-Computer Interaction and usability principles participated in the heuristic evaluation. The experts independently analyzed the selected banking portals.

General Users:

Individuals who regularly use online banking portals participated in the usability survey. A convenience sampling method was used, targeting users from different age groups and occupational backgrounds who were willing to participate voluntarily. A total of 46 participants responded to the survey.

4.5 MEASURED USABILITY METRICS

- Learnability: Ease with which new users understood how to navigate the portal
- Efficiency: Time and effort required to complete banking transactions
- Error Frequency: Occurrence of technical and input-related errors
- Cognitive Load: Amount of information users needed to remember during authentication
- User Satisfaction: Comfort and emotional response during banking interactions
- Stress Level: Anxiety experienced during financial operations

4.6 ETHICAL CONSIDERATIONS

- Participation was voluntary, and informed consent was obtained from all survey participants
- No personally identifiable financial or banking data were collected
- Survey responses were anonymized and used solely for research purposes
- The study focused on interaction behavior and user experience without recording sensitive information

4.7 DATA ANALYSIS APPROACH

Heuristic evaluation findings were analyzed qualitatively by grouping similar usability violations. Survey responses were summarized using descriptive statistics and visualized using graphs to identify common patterns and trends. Task walkthrough observations were used to support and contextualize both expert findings and user feedback.

4.8 CASE STUDY: STATE BANK OF INDIA (SBI) ONLINE BANKING PORTAL

To illustrate the practical application of the proposed methodology, the State Bank of India (SBI) online banking portal (onlinesbi.sbi) was selected as a representative example of a large-scale public banking

system. SBI is India's largest public sector bank with over 500 million account holders, making its online portal one of the most widely used banking interfaces in the country.

4.8.1 HEURISTIC EVALUATION FINDINGS

The heuristic evaluation identified several significant usability issues in the SBI online banking portal. A prominent issue was the multi-page login process requiring users to enter username on one screen, followed by CAPTCHA, then password, and finally OTP — a four-step authentication flow that imposes extremely high cognitive load and introduces multiple potential failure points. During peak banking hours, the portal frequently displayed generic error messages such as 'Authentication Failed' or 'Service Temporarily Unavailable' without providing any actionable guidance or estimated wait times.

The portal's transaction pages showed poor visibility of system status. After initiating a fund transfer, users frequently received no loading indicator and were uncertain whether to wait or resubmit, often resulting in duplicate transaction attempts. Session timeouts of 5 minutes were not adequately communicated, causing users to lose all entered data without warning.

4.8.2 TASK WALKTHROUGH OBSERVATIONS

In the task walkthrough analysis, usability breakdowns were identified at multiple stages. During login, the CAPTCHA implementation was frequently inaccessible to visually impaired users and distorted to the point of being unreadable for sighted users, requiring multiple refresh attempts. During fund transfers, the beneficiary addition process required navigation through five separate screens with inconsistent back-navigation behavior, causing users to frequently lose progress. The download of account statements required navigating through non-intuitive menu hierarchies, with the option buried under multiple sub-menus.

4.8.3 USER SURVEY INSIGHTS

Results from user surveys confirmed these expert observations. A large majority of respondents reported experiencing session timeouts during transactions, confusing error messages during failed transfers, and frustration with the multi-step authentication process. Many users reported heightened anxiety when transactions appeared to process but provided no clear confirmation, forcing them to check their balance separately to verify whether the transfer was successful.

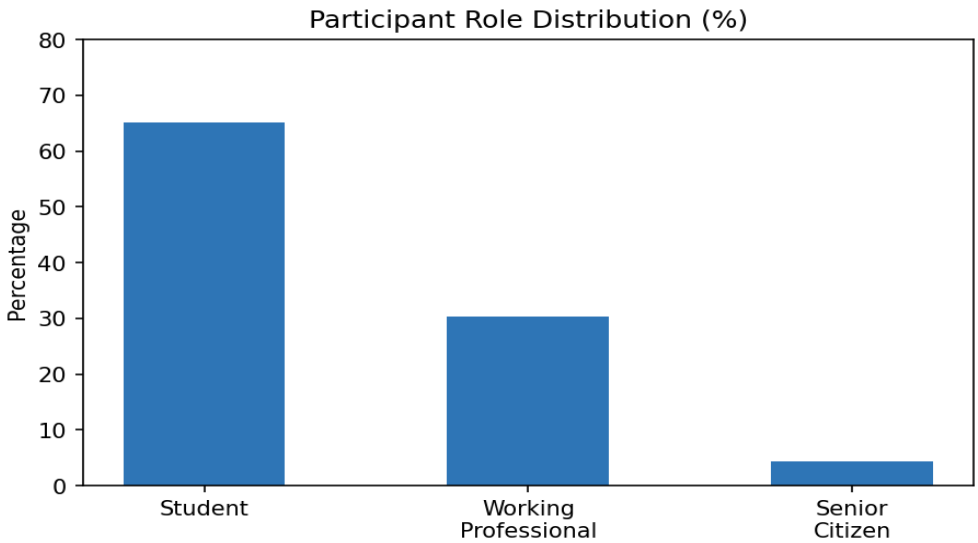
4.8.4 SUMMARY OF CASE STUDY FINDINGS

The SBI online banking portal exemplifies common usability and user experience challenges faced by large public banking institutions. The primary issues relate to excessive cognitive load during authentication, insufficient system feedback during transactions, poor error communication, and limited support for user recovery. These findings reinforce the critical importance of adherence to established Human-Computer Interaction principles to improve user experience in high-stakes financial interactions.

5. RESULTS

5.1 Participant Demographics

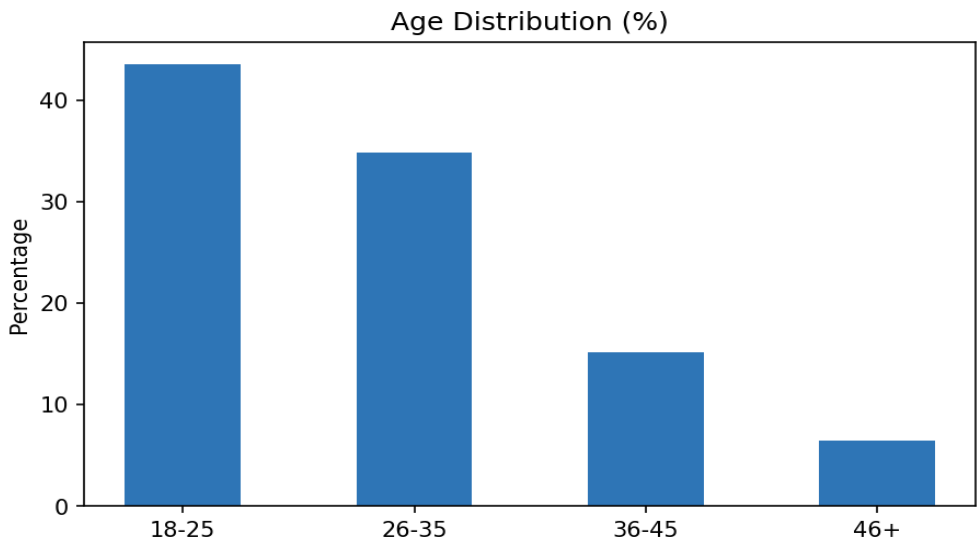
Graph 1: Role Distribution



Graph 1: Role Distribution

A total of 46 participants responded to the survey. The largest group consisted of working professionals (65.2%), followed by students (30.4%), and a small proportion of senior citizens (4.4%). This distribution ensures that findings reflect the experiences of the primary demographic that relies on online banking portals for daily financial operations.

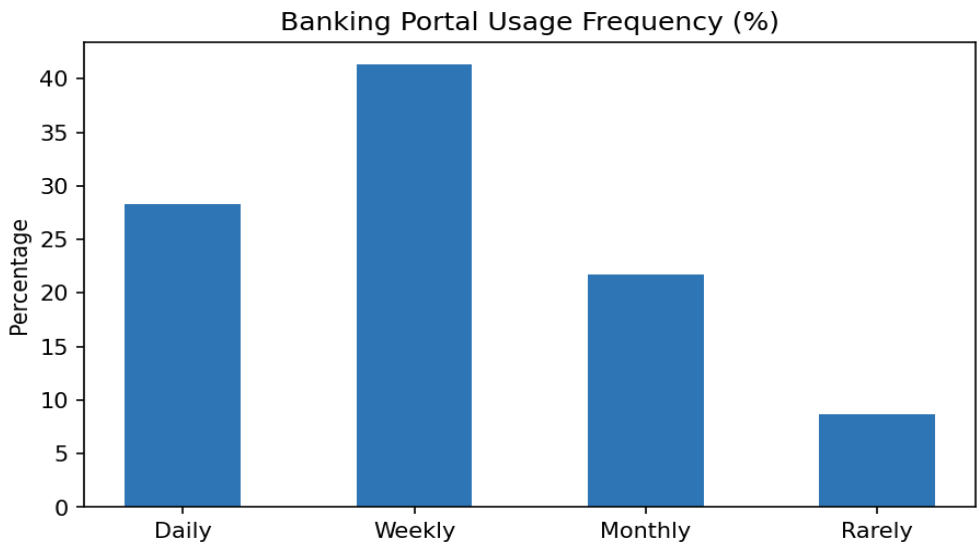
Graph 2: Age Distribution



Graph 2: Age Distribution

The age distribution shows that the 18-25 age group formed the largest segment (43.5%), followed by the 26-35 group (34.8%). The presence of participants above 36 years (21.7%) is particularly significant as older users tend to face greater challenges with recall-heavy banking interfaces.

5.2 Usage Patterns



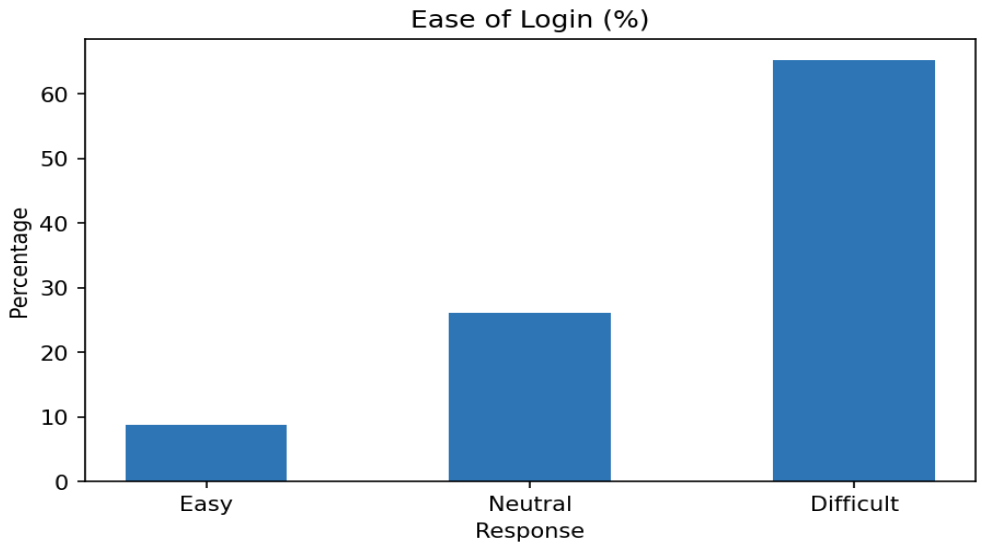
Graph 3: Usage Frequency

The results indicate that 41.3% of users access banking portals weekly, and 28.3% use them daily, highlighting the critical and frequent nature of banking portal usage. This high frequency of use makes

usability issues especially impactful, as users encounter problems repeatedly in their daily financial activities.

5.3 Usability Experience

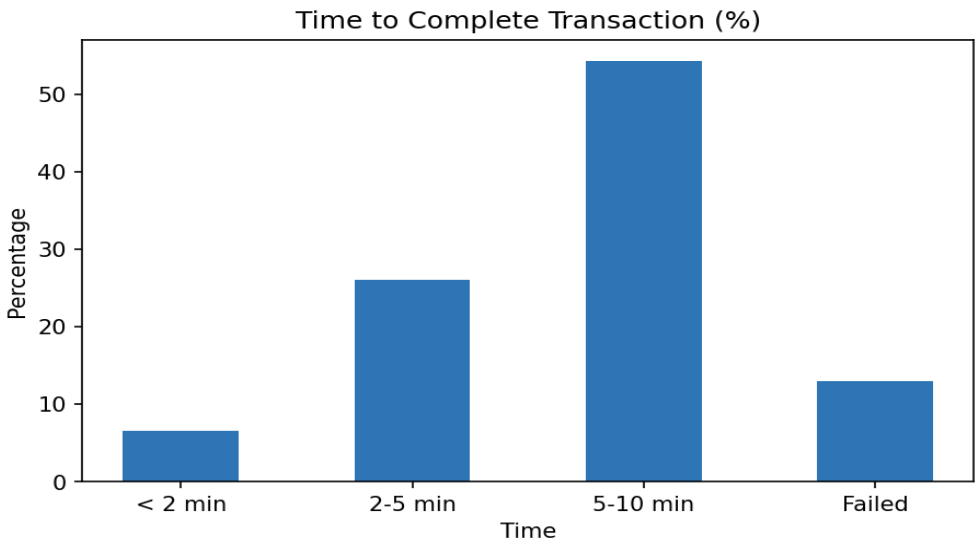
• Login Difficulty



Graph 4: Login Ease

A significant proportion of users (65.2%) reported that logging into the banking portal was difficult, while only 8.7% found it easy. This starkly indicates poor usability in the authentication interface, particularly the multi-step login process requiring users to recall multiple credentials.

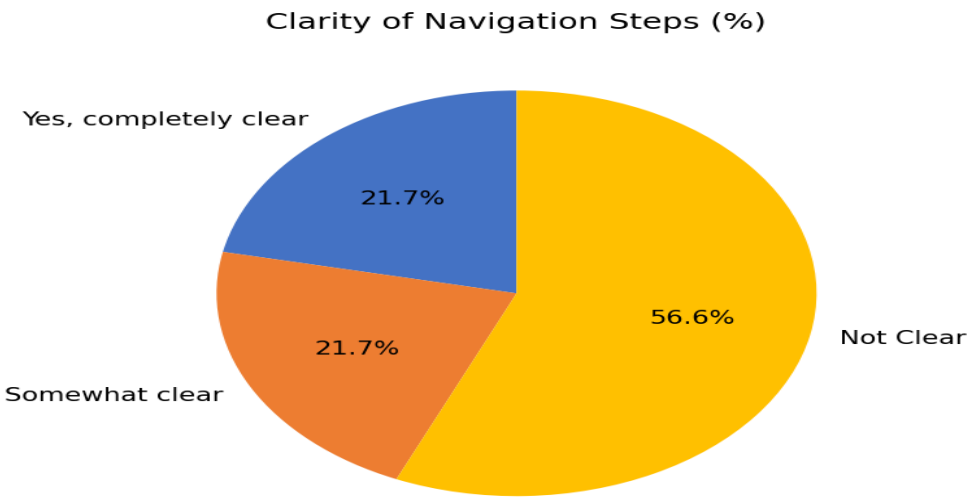
• Time to Complete Transaction



Graph 5: Time to Complete Transaction

Most users (54.3%) reported that completing a typical banking transaction took 5-10 minutes, while 26.1% required 2-5 minutes. Critically, 13% of users were unable to complete transactions successfully, indicating significant performance and usability issues. These extended transaction times reflect excessive cognitive load and poor navigation design.

• Clarity of Navigation Steps

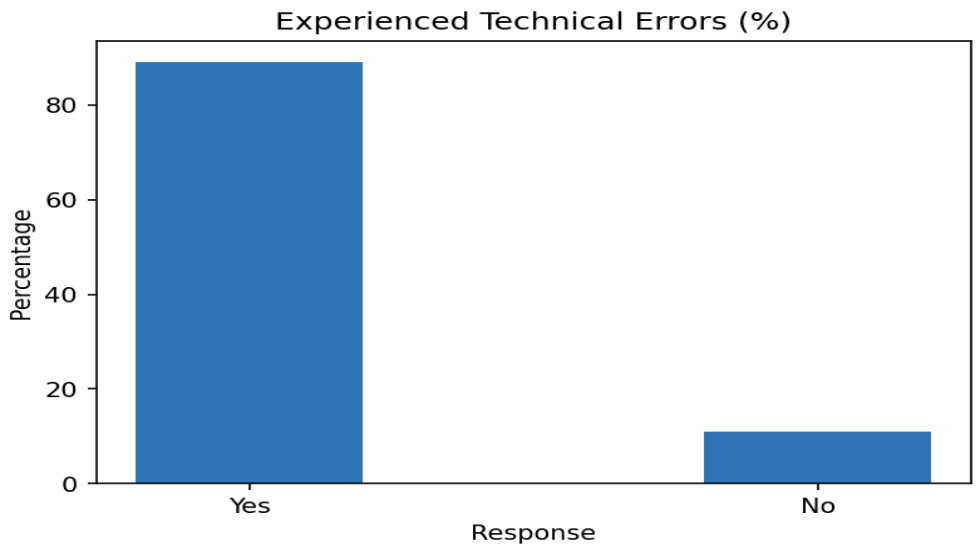


Graph 6: Clarity of Navigation Steps

Only 21.7% of users found the navigation process completely clear, while the majority (56.5%) reported that it was not clear at all. This suggests a severe lack of proper user guidance and poor information architecture in banking portal design, directly contributing to user errors and frustration.

5.4 Error Handling & Feedback

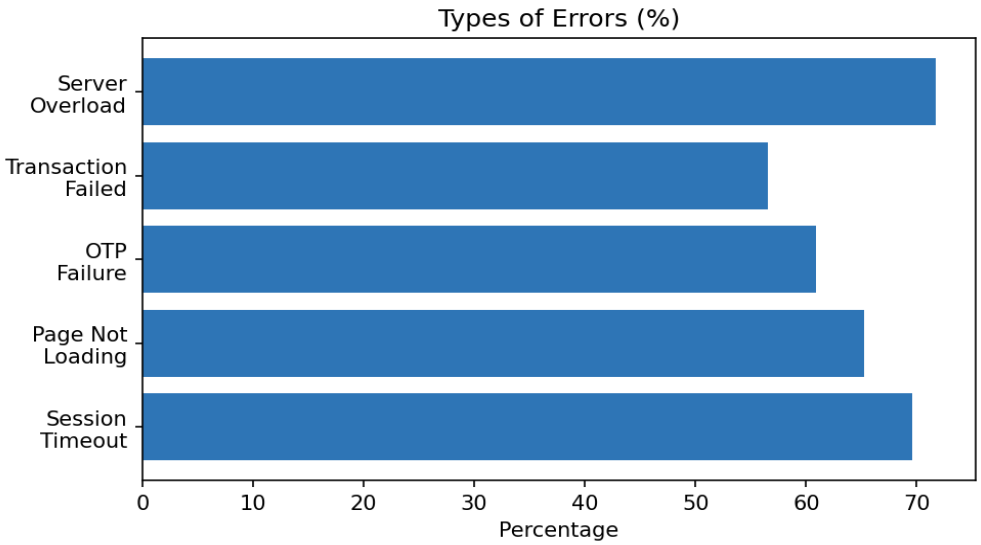
• Technical Errors



Graph 6: Technical Errors

A very high percentage (89.1%) of users experienced technical errors while using banking portals, indicating serious system reliability and usability issues. This near-universal experience of errors represents a critical failure in error prevention — one of Nielsen's core usability heuristics.

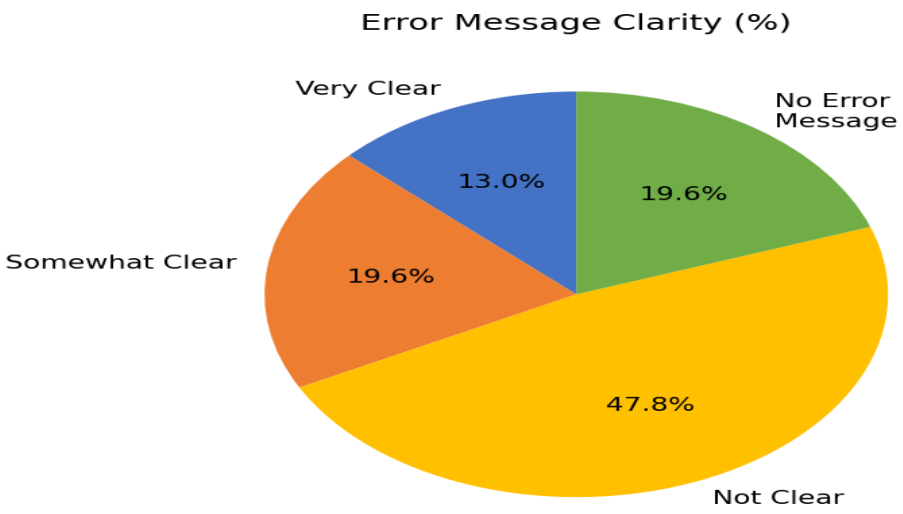
• Types of Errors



Graph 7: Types of Errors

The most common issues reported were: Server overload (71.7%), Session timeout (69.6%), Page not loading (65.2%), OTP failure (60.9%), and Transaction failed (56.5%). These findings highlight both system performance limitations and poor error recovery design in banking portals.

• Error Message Clarity

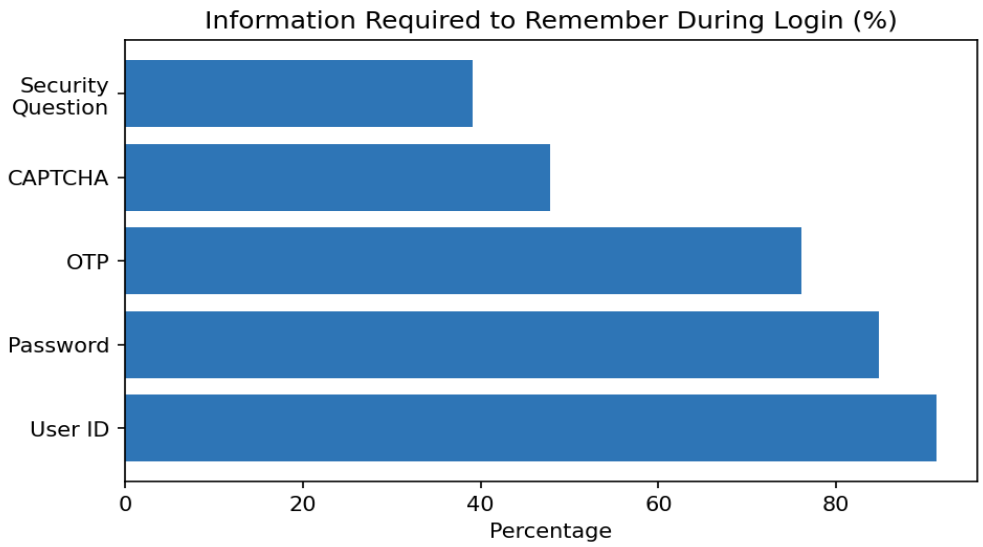


Graph 9: Error Message Clarity

Only 13% of users found error messages very clear, while 47.8% reported them as unclear or not appearing at all. This indicates critically poor feedback design — a violation of both Nielsen's 'Help users recognize, diagnose, and recover from errors' heuristic and Norman's principle of meaningful feedback.

5.5 Cognitive Load & Memory

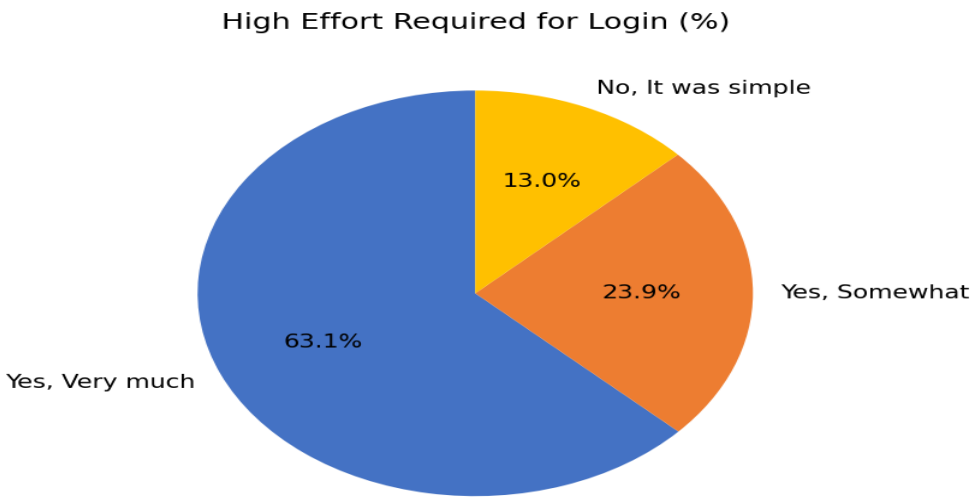
• Information to Remember



Graph 10: Information to Remember

Users were required to remember multiple pieces of information: User ID (91.3%), Password (84.8%), OTP (76.1%), CAPTCHA (47.8%), and Security Question answers (39.1%). This demonstrates an extremely high cognitive load, as users must simultaneously recall multiple credentials with different formats, validity periods, and policies — a direct violation of the recognition-over-recall principle.

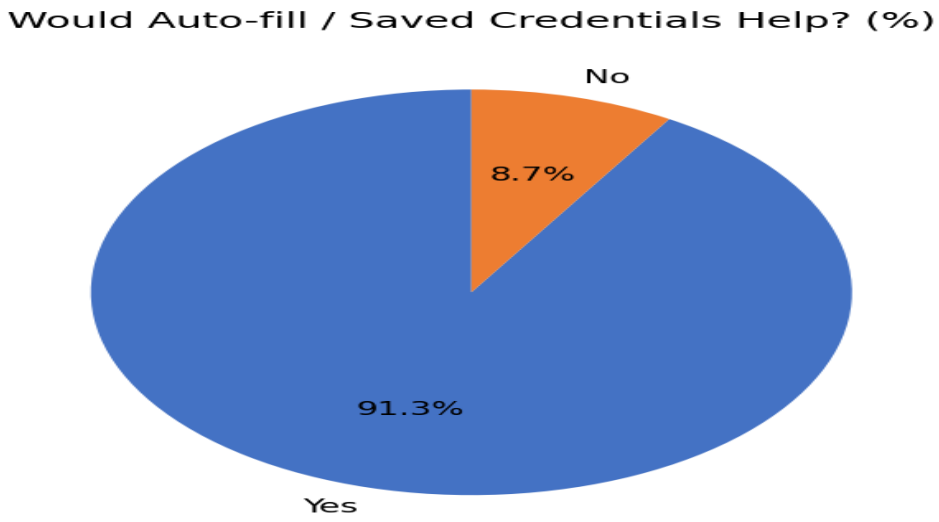
• Effort Required



Graph 11: Effort Required

A majority of users (63%) reported that the login and transaction process required very high effort, confirming the presence of poorly designed recall-based interaction throughout banking portals. Only 13% found the process simple, indicating that usability improvements are urgently needed.

• **Saved Credentials Suggestion**

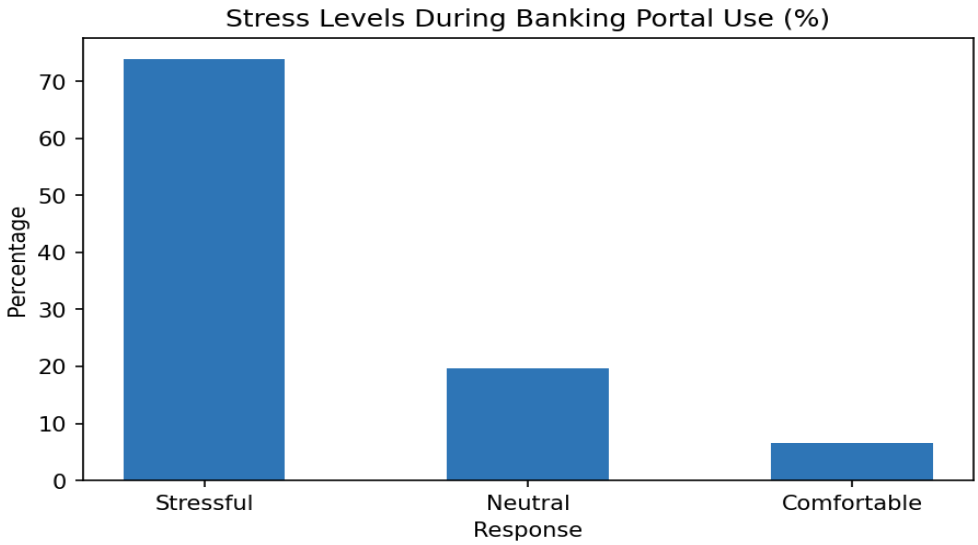


Graph 12: Saved Credentials / Auto-fill Suggestion

An overwhelming 91.3% of users agreed that features such as saved credentials, auto-fill, and biometric authentication would significantly reduce difficulty, indicating a strong user need for recognition-based alternatives to the current recall-heavy authentication systems.

5.6 Stress & Satisfaction

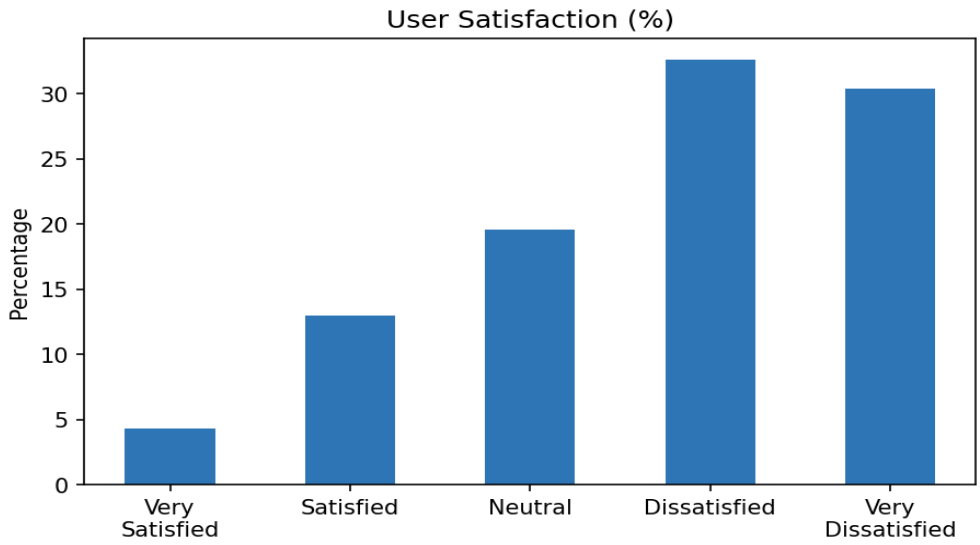
• **Stress Level**



Graph 13: Stress Levels

A large proportion (73.9%) of users reported that using banking portals was stressful, with only 6.5% reporting the experience as comfortable. This high stress rate demonstrates the severe emotional impact of poor system design in high-stakes financial interactions, where usability failures directly translate to user anxiety and distress.

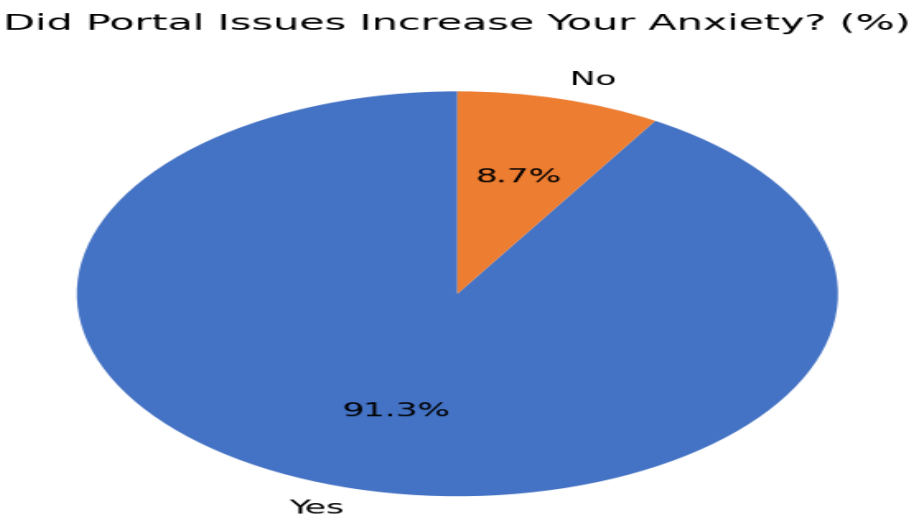
• Satisfaction



Graph 10: Satisfaction

The results show critically low satisfaction levels: Very dissatisfied (30.4%), Dissatisfied (32.6%), and Neutral (19.6%). Only a combined 17.3% of users reported any level of satisfaction. This overwhelming dissatisfaction indicates a systemic failure in user-centered design across banking portals.

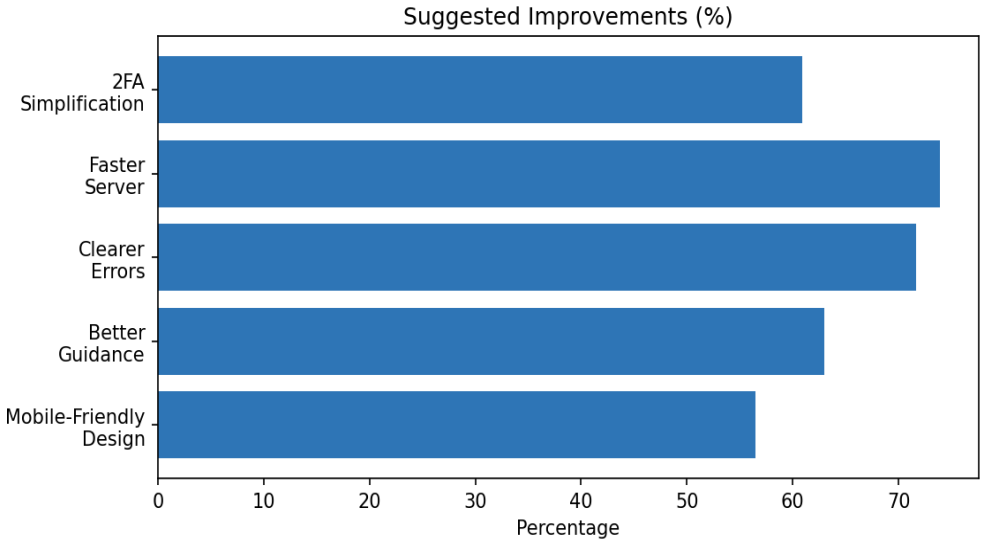
• Anxiety Impact



Graph 15: Anxiety Impact

A significant 91.3% of users reported that banking portal issues increased their anxiety levels, confirming a strong relationship between usability problems and emotional stress in financial interactions. This finding is particularly concerning given the high-stakes nature of banking operations.

5.7 Suggested Improvements



Graph 16: Suggested Improvements

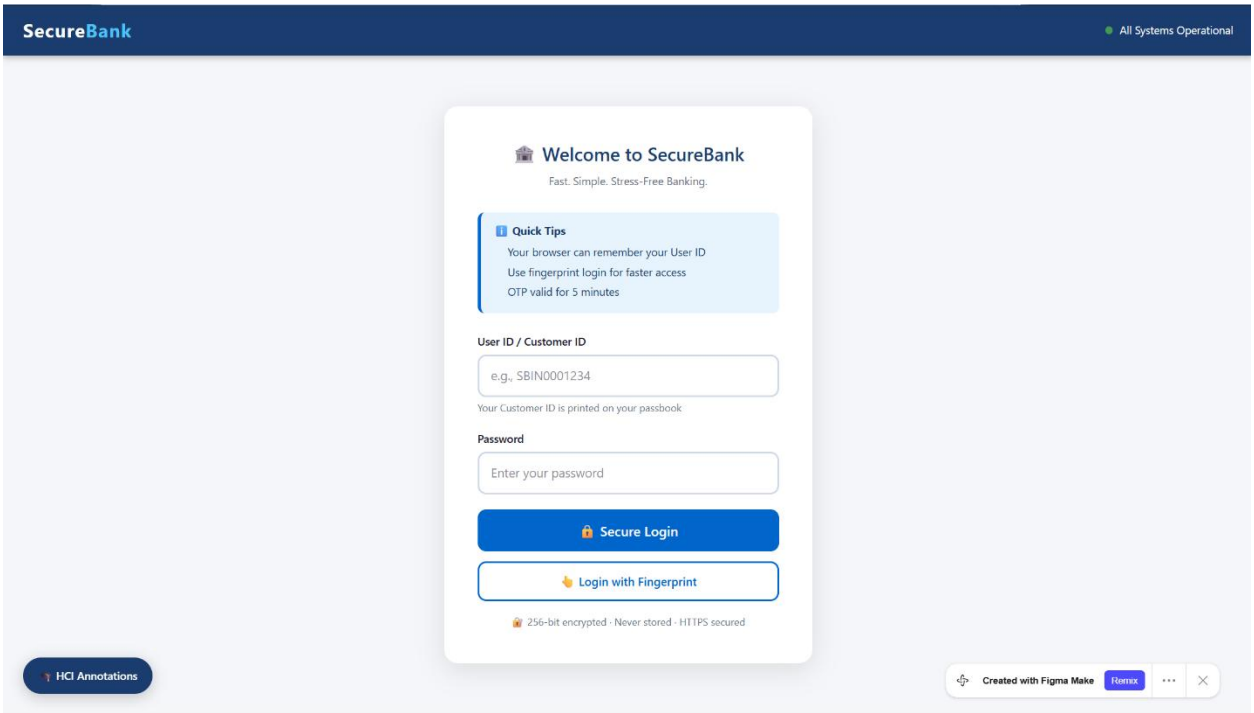
The most suggested improvements include: Faster loading and server stability (73.9%), Clearer error messages (71.7%), Better user guidance and instructions (63%), Simplified 2FA/authentication (60.9%),

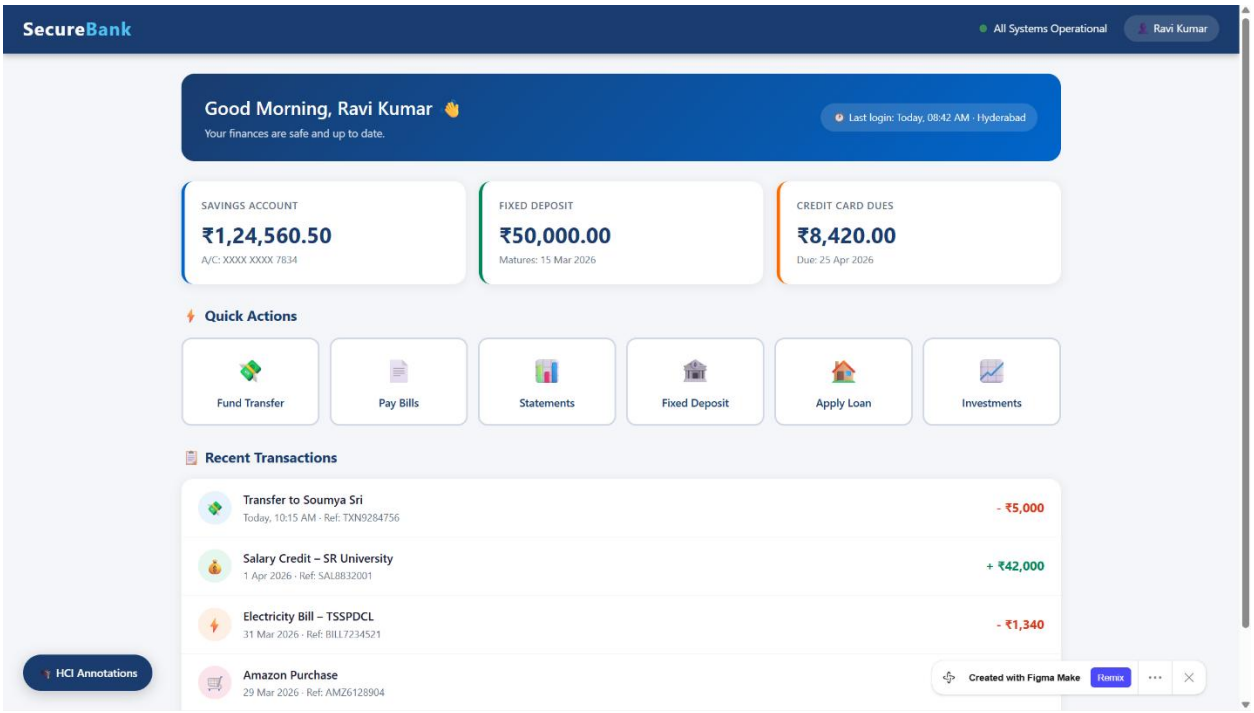
and Mobile-friendly design (56.5%). These results emphasize the need for both technical infrastructure improvements and fundamental usability redesign grounded in HCI principles.

6. PROPOSED PROTOTYPE

Figma Prototype Link:

https://www.figma.com/community/file/1622904601872857719




SecureBank
All Systems Operational
Ravi Kumar
Good Morning, Ravi Kumar
Your finances are safe and up to date.
Last login: Today, 08:42 AM · Hyderabad
SAVINGS ACCOUNT
₹1,24,560.50
A/C: XXXX XXXX 7834
FIXED DEPOSIT
₹50,000.00
Matures: 15 Mar 2026
CREDIT CARD DUES
₹8,420.00
Due: 25 Apr 2026
Quick Actions
Fund Transfer
Pay Bills
Statements
Fixed Deposit
Apply Loan
Investments
Recent Transactions
Transfer to Soumya Sri
Today, 10:15 AM · Ref: TXN9284756
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Salary Credit – SR University
1 Apr 2026 · Ref: SAL8832001
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31 Mar 2026 · Ref: BILL7234521
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HCI Annotations
Created with Figma Make
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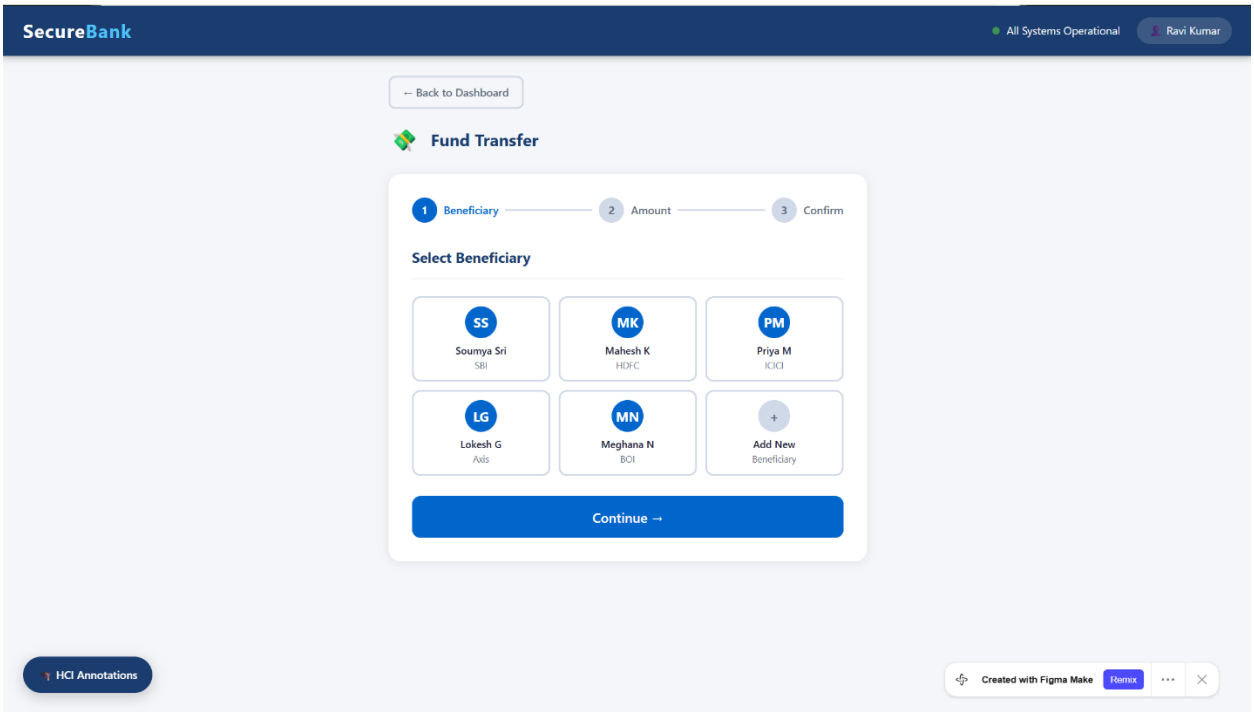
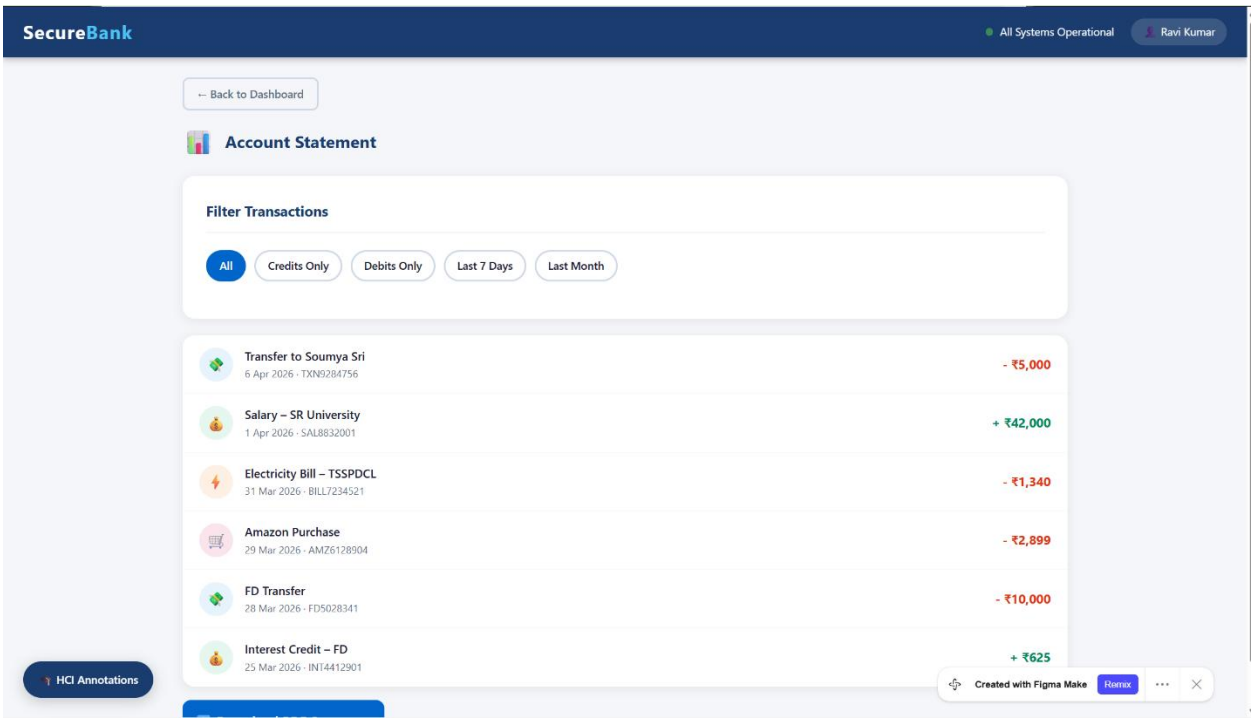

SecureBank
All Systems Operational
Ravi Kumar
← Back to Dashboard
Fund Transfer
1 Beneficiary
2 Amount
3 Confirm
Select Beneficiary
SS
Soumya Sri
SBI
MK
Mahesh K
HDFC
PM
Priya M
ICICI
LG
Lokesh G
Axis
MN
Meghana N
BOI
+
Add New
Beneficiary
Continue →
HCI Annotations
Created with Figma Make
Remix



To overcome the usability and user experience problems identified in existing online banking portals, a user-centered design prototype has been proposed. This design has been created to significantly reduce cognitive load during authentication, provide better and more timely system feedback during financial operations, and increase overall user satisfaction and trust in digital banking systems.

The proposed system provides a clean, minimal interface focused on clarity, efficiency, and usability. Unlike existing banking portals that heavily rely on recall-based multi-step authentication and provide poor transaction feedback, this design has been implemented following established Human-Computer Interaction guidelines including visibility of system status, consistency of interaction patterns, and prevention of costly errors.

Key Features of the Proposed Prototype:

Simplified Authentication Flow:

The login process has been redesigned to minimize friction while maintaining security. Rather than four separate authentication screens, the prototype consolidates authentication into a single screen with progressive disclosure. Biometric authentication support (fingerprint/face recognition) is offered as a recognition-based alternative to the current recall-heavy process. CAPTCHA has been replaced with invisible bot detection, eliminating one of the most frustrating friction points.

Real-Time Transaction Feedback:
Every user action triggers immediate, clear feedback. Fund transfers display a progress indicator with estimated completion time. Successful transactions produce a prominent confirmation screen with transaction reference number. Failed transactions provide specific, actionable error messages explaining exactly what went wrong and how to resolve it — replacing the generic 'Transaction Failed' messages common in current portals.

Session Management:
Session timeout warnings are displayed prominently 2 minutes before expiry, with a simple one-click extension option. User data entered in forms is preserved across session extensions, eliminating the frustrating loss of entered information that currently drives users to restart entire transaction flows.

Intelligent Error Prevention:
The prototype implements real-time input validation that highlights errors as users type rather than after form submission. Beneficiary account numbers are validated against bank format rules before OTP is triggered. Transaction amounts above unusual thresholds trigger a confirmation step rather than a failure after processing.

Accessibility and Inclusive Design:
Font sizes are adjustable. Color schemes meet WCAG 2.1 AA contrast requirements. All interactive elements include appropriate ARIA labels for screen reader compatibility. The interface is fully responsive and optimized for mobile use, addressing the 56.5% of users who requested mobile-friendly design improvements.

Trust and Security Indicators:
Security status is visually communicated through a persistent connection indicator. Transaction limits are clearly displayed. Users can view their last login timestamp and location immediately after authentication, building trust through transparency.

In summary, the proposed prototype demonstrates how adherence to HCI principles can transform a frustrating, anxiety-inducing banking portal into an efficient, confidence-inspiring financial interface that serves users of all ages and technical backgrounds.

7. DESIGN JUSTIFICATION USING HCI PRINCIPLES

The design of the proposed banking portal prototype is significantly informed by established Human-Computer Interaction principles from Nielsen, Norman, and Shneiderman.

Firstly, visibility of system status is maintained through real-time transaction progress indicators, session timeout warnings, and persistent security status indicators. This removes the ambiguity that currently causes users to submit duplicate transactions or abandon banking sessions out of uncertainty.

Match between system and real world is achieved by using familiar banking terminology, logical transaction flows that mirror physical banking operations, and currency formatting consistent with user expectations. Error messages use plain language rather than technical codes.

User control and freedom is ensured through the preservation of form data across session extensions, clear undo options for non-executed transactions, and explicit cancel buttons at every stage of transaction flows — giving users confidence to explore without fear of irreversible consequences.

Error prevention addresses the most critical usability gap identified in the study. Real-time input validation, bank format verification for account numbers, and confirmation dialogs for large transactions prevent errors before they occur rather than requiring users to recover after failure.

Recognition over recall is addressed through saved beneficiaries with recognizable nicknames, biometric authentication alternatives, recent transaction history for quick reference, and auto-fill of previously used payment details. This directly reduces the cognitive load that was identified as the primary driver of user stress.

Flexibility and efficiency of use is supported through quick transfer shortcuts for frequent beneficiaries, customizable dashboard with most-used services, and keyboard shortcuts for power users — while maintaining a clean, simple interface for novice users.

Aesthetic and minimalist design principles are applied by removing decorative elements that compete with functional content, using whitespace to reduce visual cognitive load, and presenting only the information essential to the current transaction step.

The proposed system therefore addresses the various shortcomings identified through heuristic evaluation and user surveys, creating a banking experience that is both secure and genuinely usable for India's diverse banking population.

8. CONCLUSION

This study aimed to investigate the usability and user experience issues of online banking portals using the principles of Human-Computer Interaction. The results of this study, based on both heuristic evaluation and user survey results with 46 participants, demonstrated that existing banking systems have significant usability issues that substantially affect user experience, trust, and financial inclusion.

The results showed that the vast majority of users (65.2%) have difficulty with the login process, nearly all users (89.1%) encounter technical errors, and an overwhelming 91.3% experience increased anxiety due to portal issues. These findings confirm that existing banking portals systematically violate core HCI principles including visibility of system status, recognition over recall, error prevention, and meaningful feedback.

The study also revealed that users face extreme cognitive loads in remembering multiple credentials including User IDs, passwords, OTPs, CAPTCHAs, and security questions — all with different formats, expiry periods, and policies. This recall-heavy interaction design, combined with poor error recovery mechanisms, directly contributes to elevated error rates and user frustration during high-stakes financial operations.

Most users expressed high levels of stress (73.9%) and low satisfaction (63% dissatisfied or very dissatisfied) when using banking portal services. More than 90% of users confirmed that portal issues increased their anxiety, demonstrating the profound relationship between usability design quality and emotional wellbeing in financial interactions.

The study highlights key areas for improvement: simplified authentication flows, real-time transaction feedback, proactive session management, intelligent error prevention, and recognition-based alternatives to recall-heavy interactions. These improvements are grounded in established HCI principles and can significantly improve user satisfaction, reduce transaction abandonment rates, and advance digital financial inclusion.

In conclusion, this study clearly demonstrates that online banking portals require urgent attention in terms of user-centered design practices. By applying established HCI principles, banking institutions can transform anxiety-inducing financial interfaces into efficient, trustworthy, and inclusive digital banking experiences that serve all users effectively.

9. REFERENCES

[1] Nielsen, J. (1994). Usability Engineering. Morgan Kaufmann.

[2] Norman, D. A. (2013). The Design of Everyday Things: Revised and Expanded Edition. Basic Books.

[3] Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., Elmqvist, N., & Diakopoulos, N. (2016). Designing the User Interface: Strategies for Effective Human-Computer Interaction (6th ed.). Pearson.

[4] Lazar, J., Feng, J. H., & Hochheiser, H. (2017). Research Methods in Human-Computer Interaction (2nd ed.). Morgan Kaufmann.

[5] Zhang, D., & Adipat, B. (2005). Challenges, Methodologies, and Issues in the Usability Testing of Mobile Applications. International Journal of Human-Computer Interaction, 18(3), 293-308.

[6] ISO 9241-210. (2019). Ergonomics of Human-System Interaction — Human-Centred Design for Interactive Systems. International Organization for Standardization.

[7] IEEE. (2022). Cognitive Load in Digital Financial Services. IEEE Transactions on Human-Machine Systems.

[8] ACM. (2023). Trust and Usability in Online Banking Interfaces. ACM CHI Conference Proceedings.

[9] Preece, J., Rogers, Y., & Sharp, H. (2019). Interaction Design: Beyond Human-Computer Interaction (5th ed.). Wiley.

[10] Reserve Bank of India. (2024). Digital Payment Trends in India. RBI Annual Report.

[11] State Bank of India. (2025). SBI Online Banking User Statistics. SBI Annual Digital Report.

[12] Google Forms Survey Data. (2026). Survey on User Experience of Online Banking Portals (n = 46).

10. GITHUB LINK

GitHub Repository: https://github.com/MRaviKumar1/Online-Banking-Portal-project-

11. FIGMA PROTOTYPE LINK

Figma Prototype: https://www.figma.com/community/file/1622904601872857719